

1 Introduction

Satellites orbits can be described by the three laws proposed by Johannes Kepler to describe planetary motion.

- **Kepler's First Law:**

The orbit of a planet is an ellipse with the Sun at one of the two foci, i.e., a satellite orbit is an ellipse with the Earth's center of mass in one of the foci.

- **Kepler's Second Law:**

A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time. Consequently, a satellites revolve with constant areal velocity.

- **Kepler's Third Law:**

The relation between the square of the period, T and the cube of the semimajor axis a is constant for all planets. Meaning, that two satellites with same a will have the same T even if the eccentricities are different.

$$T^2 = \frac{4\pi^2}{GM}a^3$$

However, this three laws would be true if no other forces than the Earth's gravity were affecting the satellites, but in reality, there exists different factors that affect the satellites motion.

But there are methods that will allowed us to compute satellites' positions, this methods will parametrize the orbits and will take the perturbations in consideration while computing the positions. It is important to take into consideration that the forces that produce the perturbations are very difficult to model, meaning that the positions can be predicted but with a limited accuracy.

Computing satellite coordinates involves determining the precise position of a satellite in its orbit using various orbital parameters and mathematical models. However, computing these coordinates presents several challenges. Satellites are affected by various forces, including the gravitational pull of the Earth, Moon, and Sun, as well as solar radiation pressure. These perturbations can cause deviations in the satellite's orbit, making accurate computation difficult. At the same time, the high velocities and varying gravitational potentials experienced by satellites require relativistic corrections to ensure accurate timing and positioning. Atmospheric conditions, such as ionospheric and tropospheric delays, can affect the accuracy of signals received from satellites. While the quality of the data is responsible for the accuracy of the coordinate computation. [5]

Computing satellite coordinates from broadcast ephemerides involves determining the precise position of a satellite in its orbit using the ephemeris data transmitted by the satellite.

Satellite ephemeris data is the data that describes the precise orbital parameters of a satellite at a given time. This data includes various orbital parameters that describe the satellite's trajectory and is essential for accurate navigation and timing. Ephemeris data typically includes information such as the satellite's orbital elements, clock corrections, and other parameters necessary for accurate navigation and timing. The orbital elements of the ephemeris are the parameters that define the shape and orientation of the satellite's orbit, including semimajor axis, eccentricity, inclination, argument of perigee, and longitude of ascending node. The clock correction parameters are the adjustments to the satellite's on-board clock to account for relativistic effects and other factors.

Finally, the perturbation parameters are the corrections for the gravitational effects of the Earth, Moon, and Sun, as well as other forces affecting the satellite's orbit.[12]

The ephemeris data transmitted by the satellites is obtained from the navigation message files, which are in RINEX format. A RINEX (Receiver Independent Exchange Format) file is a standardized data format used to store and exchange GNSS (Global Navigation Satellite System) data. RINEX files are essential for processing and analyzing satellite navigation data from various GNSS systems such as GPS, Galileo, GLONASS, and BeiDou. [3]

Accurate satellite positioning is critical for Global Navigation Satellite Systems (GNSS) such as GPS, Galileo, and others. These systems rely on accurate satellite coordinates to provide precise location information to users on Earth. Errors in satellite positioning can lead to significant inaccuracies in navigation and timing, affecting everything from personal navigation devices to critical infrastructure systems.

However, satellite clock errors directly affect the accuracy of the computed satellite coordinates; at the same time, even small inaccuracies in the satellite clock can lead to significant errors in the computed position of the GNSS receiver. Satellite clock errors refer to inaccuracies in the timekeeping of the atomic clocks on board GNSS satellites. These errors can be caused by clock drift, noise, and relativistic effects. Despite the high accuracy of atomic clocks, small deviations can occur that affect the accuracy of the satellite's time signal. These errors can lead to positioning errors, as discussed earlier, but also to timing errors when these satellite clock errors affect the synchronization of time signals, which is critical for tasks that require precise timing, such as telecommunications and financial transactions. Inaccurate clock data can make it more difficult for GNSS receivers to acquire and track satellite signals, affecting the overall performance of the navigation system. [14]

There are some mitigation strategies for these clock errors, for example, GNSS satellite navigation messages include correction parameters to help receivers adjust for known clock errors and improve accuracy. At the same time, one of the tasks of the GNSS ground control segment is to continuously monitor satellite clocks and update the broadcast ephemeris with corrections. Augmentation systems provide additional corrections for satellite clock errors, improving positioning accuracy.

2 Methodology

The process of determining the position of a GPS satellite at a given epoch involves the use of broadcast ephemeris parameters and solving Kepler's equations iteratively. Additionally, satellite clock corrections are computed to improve positioning accuracy.

The process is described in [9]. And it consists in seven different steps that from broadcast ephemeris will compute GPS satellites and satellite clock correction

1. Computation of transmission time.
2. Computation of satellite clock correction.
3. Computation of system transmission time.
4. Computation of eccentric anomaly.
5. Recalculation of transmission time using relativistic corrections.
6. Repetition of previous steps for precise timing.
7. Computation of satellite's Cartesian coordinates.

To perform the process to compute the coordinates of the GPS satellites (X^s, Y^s, Z^s) and the satellite clock correction (δt_{L1}^s) using broadcast ephemeris, we have create a Matlab file that given the pseudoranges with the P-code P1, and the RINEX file with the data for the different satellites, it performs all the computations that will be explained bellow.

To facilitate the process, we have create three functions for the steps that are repeated in the process. We have define a function **step4**, **step5** and **definetk**. The first two functions that collect the computations previously explained for step 4 and step 5, making easier the computation of step 6. The function **definetk** is called when doing step 4, and refers to the process to compute the ephemeris reference epoch t_k .

To make the code easier to understand, the computations of step 7, have been introduced in a function called **step7**.

For the task of comparing the satellites' coordinates, we have take from the file **igs12561.sp3** the coordinates corresponding to the epoch 2004 - 2 - 2 01:00:00. To check our computed coordinates, we have compute the error as the difference between the precise satellite positions and the computed coordinates.

Now that the structure of the method use is explained, we will introduce the details of the calculations in the seven steps.

1. Computation of transmission time.

The transmission time measured by the clock of the satellite s , $\tilde{t}_s$, is computed by correcting the receiver's observation epoch $\tilde{t}_A$ using the pseudorange $P_a^s(\tilde{t}_A)$:

$$\tilde{t}_s = \tilde{t}_A - \frac{P_a^s(\tilde{t}_A)}{c}$$

where $\tilde{t}_A$ is the receiver's GPS time (in seconds of the week), c is the speed of light (299792458 m/s), and $P_a^s(\tilde{t}_A)$ is the measured pseudorange in meters.

In this case we are going to use only the pseudoranges classified as P1, which a classification depending of the P-code used. To obtain this pseudoranges, we should take from the RINEX file that contains all the observations, only the ones that corresponds with the P1 pseudoranges, in this case the fourth column of numerical data.

2. Computation of satellite clock correction.

The satellite clock correction δt_{L1}^s is necessary when computing the receivers position, and it represents the correction of an error in the clock of the satellite. It is computed as:

$$\delta t_{L1}^s = \Delta t_{sv} - TGD$$

where TGD is the timing group delay in seconds, this parameter is used in Global Navigation Satellite System (GNSS) data to account for the delay caused by the satellite's signal traveling through the ionosphere. This delay affects the accuracy of the positioning data, and TGD is used to correct for this delay in the calculations.[13] And Δt_{sv} is the satellite clock offset that accounts for drift in the satellite clock,

$$\Delta t_{sv} = a_{f0} + a_{f1}(t_s - t_{oc}) + a_{f2}(t_s - t_{oc})^2$$

where t_{oc} is the clock reference time according to the RINEX file of the broadcast ephemeris and a_{f0} , a_{f1} , and a_{f2} are the satellite clock bias, drift, and drift rate from the broadcast ephemeris, also given in the RINEX file of the broadcast ephemeris.

3. Computation of system transmission time.

The corrected system transmission time is:

$$t_s = \tilde{t}_s - \delta t_{L1}^s$$

Meaning that it is the nominal time of the signal transmission measured by the clock of the satellite s, corrected by the clock correction factor for the satellite.

4. Computation of eccentric anomaly.

The mean anomaly is a measure of the average angular position of a satellite in its orbit, assuming it moves at a constant speed. It is calculated based on the time elapsed since the satellite passed the perigee (the closest point to Earth in its orbit) and the mean motion (average rate of orbit). The mean anomaly helps in predicting the satellite's position at any given time. It can be computed as:

$$M_k = M_0 + n \cdot t_k$$

where, M_0 is the mean anomaly at reference time. The corrected mean motion n is given by:

$$n = n_0 + \Delta n$$

Where $n_0 = \sqrt{\frac{\mu}{A^3}}$ is the computed mean motion and A is the semi-major axis given in the RINEX file.

The time from the ephemeris reference epoch t_k is the specific time at which the satellite's ephemeris data is valid. It is computed as:

$$t_k = t_s - t_{oe}$$

$$t_k = \begin{cases} t_k - 604800, & \text{if } t_k > 302400 \\ t_k + 604800, & \text{if } t_k < -302400 \\ t_k, & \text{otherwise} \end{cases}$$

The parameter t_{oe} is the Time of Ephemeris obtained from the RINEX file. It indicates the reference time for the satellite's ephemeris data, which includes the satellite's position and velocity. This time is essential for accurately determining the satellite's position at any given moment.[13]

The eccentric anomaly E_k is an angular parameter that relates the position of a satellite in its elliptical orbit to the mean anomaly, it accounts for the elliptical shape of the orbit and is essential for precise calculations of the satellite's position. It is solved iteratively using Kepler's equation:

$$E_k = M_k + e \sin E_k$$

where e is the orbital eccentricity.

For the iterative process that help us obtain the eccentric anomaly, we set initially $E_0 = M_k$. For the following steps, we compute it as

$$E_i = M_k + e \sin E_{i-1}, i = 1, 2, 3...$$

The iteration can be stop when $|E_i - E_{i-1}| < \epsilon$, where $\epsilon = 1e - 13$.

5. Recalculation of transmission time using relativistic corrections.

The relativistic correction due to the satellite's orbit is:

$$\Delta t_r = F \cdot e \sqrt{A} \sin E_k$$

Where F is the constant $-4.442807633e - 10s/m^{1/2}$, e is the orbital eccentricity, A is the semi-major axis and E_k is the eccentric anomaly.

The relativistic correction due to a satellite's orbit accounts for the effects of both general relativity and special relativity on the satellite's clock. The rate of advance of two identical clocks, placed one in the satellite and the other on the terrestrial surface, will differ due to the difference of the gravitational potential (general relativity) and to the relative speed between them (special relativity). These corrections are necessary because the satellite is moving at high speeds and is in a different gravitational potential compared to the Earth's surface. [6]

The satellite clock offset is updated is updated by the relativistic correction:

$$\Delta t_{SV} = a_{f0} + a_{f1}(t_s - t_{oc}) + a_{f2}(t_s - t_{oc})^2 + \Delta t_r$$

Thus, the final satellite clock correction is as before:

$$\delta t_{L1}^s = \Delta t_{SV} - TGD$$

6. Repetition of previous steps for precise timing.

Since t_s has been updated for precision we should repeat steps 4 and 5. We recompute:

1. The eccentric anomaly E_k ,
2. The mean anomaly M_k ,
3. The satellite clock correction δt_{L1}^s ,
4. The corrected transmission time t_s ,
5. The time from the ephemeris reference epoch t_k .

7. Computation of satellite's Cartesian coordinates.

The true anomaly, represents the angle between the direction of the perigee (the closest point in the orbit to Earth) and the current position of the satellite, as seen from the center of the Earth, it is computed as:

$$v_k = \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin E_k}{1 - e \cos E_k} \right)$$

The argument of latitude, is an angular parameter used in orbital mechanics to describe the position of a satellite in its orbit. It combines the effects of the argument of perigee (ω , given in the RINEX file) and the true anomaly:

$$\phi_k = v_k + \omega$$

When computing satellite coordinates, several corrections are applied to account for various perturbations and ensure accurate positioning.

The argument of latitude correction (δu_k) adjusts for periodic variations in the satellite's orbit due to gravitational influences from the Earth, Moon, and Sun. [8]

$$\delta u_k = c_{us} \sin 2\phi_k + c_{uc} \cos 2\phi_k$$

Where c_{us} is the amplitude of the sine harmonic correction term applied to the argument of latitude. It accounts for periodic variations in the satellite's orbit due to gravitational influences. This correction helps in accurately determining the satellite's position in its orbit. And, c_{uc} is the amplitude of the cosine harmonic correction term applied to the argument of latitude. Similar to c_{us} , it adjusts for periodic variations but uses the cosine function. This correction is essential for precise calculations of the satellite's position. [4]

The radius correction (δr_k) accounts for deviations in the satellite's orbital radius caused by gravitational perturbations and other forces. This correction ensures that the calculated distance from the satellite to the Earth's center is accurate. [14]

$$\delta r_k = c_{rs} \sin 2\phi_k + c_{rc} \cos 2\phi_k$$

Where c_{rs} is the amplitude of the sine harmonic correction term applied to the orbit radius. It accounts for periodic variations in the satellite's orbital radius due to gravitational influences. This

correction helps in accurately determining the satellite's distance from the Earth's center. And, c_{rc} is the amplitude of the cosine harmonic correction term applied to the orbit radius. It adjusts for periodic variations but uses the cosine function. This correction is essential for precise calculations of the satellite's orbital radius. This parameters are given in the RINEX file.[4]

The inclination correction (δi_k) adjusts for changes in the satellite's orbital inclination due to gravitational influences, primarily from the Earth's equatorial bulge. This correction is crucial for maintaining the satellite's intended orbital plane. [7]

$$\delta i_k = c_{is} \sin 2\phi_k + c_{ic} \cos 2\phi_k$$

Where, c_{is} is the amplitude of the sine harmonic correction term applied to the angle of inclination. It accounts for periodic variations in the satellite's inclination due to gravitational influences. This correction helps in accurately determining the satellite's orbital plane. And, c_{ic} is the amplitude of the cosine harmonic correction term applied to the angle of inclination. It adjusts for periodic variations but uses the cosine function. This correction is essential for precise calculations of the satellite's inclination. This parameters are given in the RINEX file.[4]

Applying the previous corrections:

$$\begin{aligned} u_k &= \phi_k + \delta u_k, \\ r_k &= A(1 - e \cos E_k) + \delta r_k, \\ i_k &= i_0 + \delta i_k + IDOT \cdot t_k, \end{aligned}$$

Where they are respectively the corrected argument of latitude, the corrected radius and the corrected inclination.

From the RINEX file we obtain the IDOT, which is the Rate of Inclination Angle. It represents the rate at which the satellite's orbital inclination changes over time. This parameter is crucial for accurately predicting the satellite's position in its orbit, as it accounts for the gradual changes in inclination due to gravitational influences. And i_0 , is the Inclination Angle at the Reference Epoch. It specifies the angle between the satellite's orbital plane and the equatorial plane at the reference time (epoch). This angle is a fundamental parameter in defining the satellite's orbit and is used as a baseline for calculating the satellite's position. [11]

According to the previous computed parameters, the satellite's position in its orbital plane is:

$$x'_k = r_k \cos u_k, \quad y'_k = r_k \sin u_k$$

Transforming the coordinates to the Cartesian coordinates with the parameters of the ellipsoid WGS84, we obtain the following expressions for the coordinates:

$$\begin{aligned} x_k &= x'_k \cos \Omega_k - y'_k \sin \Omega_k \\ y_k &= x'_k \sin \Omega_k + y'_k \cos \Omega_k \\ z_k &= y'_k \sin i_k \end{aligned}$$

Where Ω_k accounts for the corrected longitude of ascending node.

$$\Omega_k = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e) t_k - \dot{\Omega}_e t_{oe}$$

This parameter refers to the perturbations affecting the satellite's orbit. This correction ensures accurate positioning by adjusting the longitude of the ascending node, which is the point where the

satellite crosses the equatorial plane from south to north. In its computation, we use the RINEX parameters Ω_0 , $\dot{\Omega}$ and $\dot{\Omega}_e$. The first one is the Longitude of the Ascending Node at the reference epoch. It represents the angle from a fixed reference direction (usually the vernal equinox) to the point where the satellite crosses the equatorial plane from south to north. $\dot{\Omega}$ is the Rate of Change of the Longitude of the Ascending Node. It accounts for the precession of the satellite's orbit due to gravitational perturbations, primarily from the Earth's oblateness. And, $\dot{\Omega}_e$ is the Rate of Earth's Rotation. This parameter is used to account for the rotation of the Earth in the calculations of the satellite's position.[10]

3 Results

The table 1 contains the computed coordinates for each satellite and the corresponding clock error. These results have been checked by comparing the data obtained for satellites 24 and 29 with those given in **Satellite24.pdf**. We have obtained a good accuracy for the computed coordinates, while for the clock correction error we see that the last decimals are a little different, this could be due to the computer memory of the decimals. At the same time, for satellite 24, we have also checked the values of each step, and we can also see the deviation in the last decimals of the computed values. Overall, the results are accurate when compared to the results given in the citation document.

Satellite	X	Y	Z	Clock error
2	-13921125.8457202	10520857.5163205	20752900.0078171	-0.000267700910515206
3	-12326721.4233817	15206582.1121522	17758735.4306721	8.25532966361938e-05
8	19782985.9943486	3245004.08849558	17378780.211023	0.000374225642453191
10	16224653.0411999	-6015356.85329839	20186062.2024433	3.87579882886296e-05
13	7552200.21749659	22602205.1271002	11771799.8848144	-3.13711368993048e-05
17	16522611.9718119	-16233816.0690998	12453389.8682165	0.000158216248933414
21	-11894220.6599127	-10766056.1061283	21172793.1290697	7.1724874454825e-05
24	23421962.7682529	-12587454.8961215	999742.383390349	1.10485271457453e-06
26	8280837.76410672	-21834331.7635575	11796587.6469314	0.000438364921417737
27	9544631.97106284	12516502.6452464	21878718.7572797	0.000904304466369752
28	23778312.6869532	11792190.0751523	-2950541.18196521	1.75614259098107e-05
29	11084174.9636596	-17544452.6845597	16816661.0643185	0.000219487940928332

Table 1: Satellite Coordinates and Clock Errors

The file **igs12561.sp3** contains the exact satellite positions, these exact coordinates are helpful to eliminate the effects of errors in the broadcast ephemeris. We have compared the coordinates given in this file for epoch 2004 - 02 - 02 01:00:00 with the coordinates computed from the broadcast satellite ephemeris data. The error measurement against the coordinates is included in the table ?? and all error measurements are in meters. From the error measurements we can see that there is a small deviation between the two coordinates, this could be due to some of the forces and effects that can affect the received signal.

Satellite	error coordinate X	error coordinate Y	error coordinate Z
2	3.2756	1.0800	-0.4401
3	-0.5451	-1.4536	-1.0596
8	-0.5311	2.6697	-1.5717
10	0.6577	2.6052	0.1347
13	-3.1938	-0.9528	0.2142
17	0.3837	1.1297	0.3415
21	0.1516	1.8717	-0.9177
24	0.6466	1.0540	3.8622
26	-0.2256	-0.3388	-1.5418
27	-1.6013	-0.1301	-0.5140
28	-1.8366	0.2649	0.0060
29	-1.2658	-3.2786	-2.5365

Table 2: Errors in Satellite Positioning (in meters)

4 Analysis and Discussion

As it was introduced in the previous section, we can see that, given the results and the parameter values of the intermediate steps, we have achieved a very high accuracy in the calculation of the coordinates and the time correction. However, if we compare the results with the precise satellite positions, we can see that there is some difference in the values, which in some cases could lead to an error of almost 4 meters in the positioning of the satellites.

This difference between the coordinates could be due to a clock error. When we calculated the clock error and compared it to the results in **Satellite24.pdf**, we saw that there is a small difference in the last decimals. Although this error in the decimals does not seem like a big change, it means that further refinement may be needed. At the same time, we must consider the existence of clock errors. Satellite clocks drift slightly due to imperfections in on-board atomic clocks and relativistic effects, and even with clock corrections transmitted in navigation messages, small residual errors remain, and poor processing of this error could lead to a large deviation in coordinates.

Achieving greater accuracy in the computation of coordinates is a very important task, since inaccurate positioning of the satellite could lead to inaccurate positioning of the receiver, which could have terrible consequences, especially in navigation or military operations.

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